RLI as Regional Relational Ontology: Boundaries and Scope

域内锁定推断作为区域关系本体论：边界与范围

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Abstract.
RLI is a regional relational ontology. Inside the locking network, relations are prior to entities. Outside the region, the language of RLI does not refer. The boundary is not a gap but an identity condition.

摘要.
RLI 是区域关系本体论。在锁定网络内，关系优先于实体。区域外，RLI 的语言不指称。边界不是缺口，是身份条件。

I. Region / 区域

Region = locking network, $\gamma > 0$.

区域 = 锁定网络，$\gamma > 0$。

Ground state $\gamma_{bg} \approx 0$ is a **limit point inside the region**, not a boundary. Locking still exists ($\gamma > 0$), only the network average vanishes.

基态 $\gamma_{bg} \approx 0$ 是**区域内的极限点**，不是边界。此时锁定仍然存在（$\gamma > 0$），只是网络平均为零。

Boundary: $\gamma = 0$**.** The conservation law $\rho + N\gamma_{bg}/\gamma = u_0^2$ becomes $0/0$ here; the equation is undefined. Language does not refer.

边界： $\gamma = 0$。 守恒式 $\rho + N\gamma_{bg}/\gamma = u_0^2$ 在此为 $0/0$，方程无定义。语言不指称。

II. Relation / 关系

A1. Measurement is the process of establishing a relational lock between two intrinsic locking structures.

A1. 测量是两个内禀锁定结构之间建立关系锁定的过程。

Relations are prior to entities. Particles, fields, spacetime are unfoldings of locking, not pre-existing entities. The intrinsic is the trace left by relations.

关系优先于实体。粒子、场、时空是锁定的展开，不是预先存在的实体。内禀 = 关系留下的痕迹。

III. Ontology Inside the Region / 区域内本体论

Derived from A1, stratified:

从 A1 推出，层级分明：

硬核

A1, three nodes, 4π double cover, $3\pi\delta = 2\pi/3$, four densities, Ω_b/Ω_m, $Q = 2/3$.

A1、三节点、4π 双覆盖、$3\pi\delta = 2\pi/3$、四条密度、Ω_b/Ω_m、$Q = 2/3$。

候选

α, $1/6$, Γ_{AA}, glueball, vacuum energy density, mass ratios, M_R.

α、$1/6$、Γ_{AA}、胶球、真空能密度、质量比、M_R。

接口

Commutation relations, chiral anomaly, coarse-graining merge rule.

对易关系、手性异常、粗粒化合并规则。

IV. Outside the Region / 区域外

Five classes below are not "unanswerable" but "language does not refer":

以下五类，不是“答不了”，是“语言不指称”：

1. $\gamma = 0$. The domain of γ is $(0, \infty)$; "$\gamma = 0$" has no referent.

   $\gamma = 0$。γ 的定义域是 $(0, \infty)$，“$\gamma = 0$”没有指称。

2. Before the break. "Before" presupposes time, which is defined only after A1.

   破之前。“之前”预设时间，时间在 A1 之后才有定义。

3. The mechanism of the break. Requires standing outside A1; RLI has no such language.

   破的机制。要求站在 A1 外面回答，而 RLI 没有外面的语言。

4. Value judgments. "Ought" is not in RLI's vocabulary.

   价值判断。“应该”不在 RLI 的语汇里。

5. Qualia. "Feeling" presupposes a first person; RLI's first person is the loop closure point, not phenomenal experience.

   Qualia。“感觉”预设第一人称，RLI 的第一人称是回路闭合位置，不是现象学体验。

V. Relationship to Other Ontologies / 与其他本体论的关系

Substance ontology. Different. Relations are prior; no independent entities are presupposed.

实体论。不同。关系优先，不预设独立实体。

Process ontology. Partial overlap. RLI has a process flavor, but the starting point is relation, not process.

过程论。部分重叠。RLI 有过程气质，但起点是关系，不是过程。

Moderate Ontic Structural Realism. Closest. Relations are prior; objects are characterized by relations but not eliminated. Difference: Moderate OSR starts from quantum mechanics; RLI starts from A1.

Moderate Ontic Structural Realism. 最接近。关系优先，对象由关系刻画，不取消对象。区别：Moderate OSR 从量子力学出发，RLI 从 A1 出发。

VI. Name / 名称

Internal code **PLI** (Pure Locking Inference). External identifier **RLI** (Regional Locking Inference).

内部代号 **PLI**（Pure Locking Inference）。对外标识 **RLI**（Regional Locking Inference）。

Chinese external name: 域内锁定推断.

中文对外名：**域内锁定推断**。

"Pure" means pure inside the region, not globally unbounded.

“Pure”指区域内纯粹，不指全局无边界。